

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Philadelphia, PA

Expected May 2028

Expected May 2027

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Aug 2026 · San Francisco (Remote)

- **Stopped a release that would have shipped to zero users** — the planned over-the-air update pointed at a runtime a dependency upgrade had invalidated: it would have deployed, reported success and reached nobody. Caught it against the live binaries' fingerprints and rerouted to a store build
- **Unblocked a population that could never have been tested** — the rollout flag was scoped by email, which cannot resolve before sign-in, so signed-out and first-time users could never receive the new experience, including the app's own login and first-class screens
- **Turned a channel that went dark whenever nobody approved a send into one that runs itself** — proposing, clearing its own guardrails and reaching **91% of targeted devices** unattended, producing the win-back's first ever revenue at **~2.7× the reach** once I proved its eligibility logic had excluded most of its audience
- **Made a children's app safe from notification injection** — anyone with a user ID could push arbitrary content until I bearer-authed four public endpoints, **zero caller changes**

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- **Kept 50+ live Unity mobile titles shipping** across a summer of releases, holding UI/UX and gameplay stable
- **Cleared 100+ QA-reported bugs into production**, in Unity prefabs and C# across a portfolio no single engineer owned
- **Kept the portfolio compliant and in the stores** through a platform requirement change, upgrading SDKs fleet-wide

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- **Shipped client-facing 0-to-1 products** leading a cross-functional team of **15+** designers and developers
- **Made design decisions once instead of per-person**, defining the end-to-end flows and reusable interface systems the team built inside

Military Interpreter, Capital Defense Command (ROK Army)

Dec 2022 – June 2024 · Seoul, South Korea

- **Held coordination together across government and defense organizations** as real-time English–Korean interpreter in high-security, multi-agency settings

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- **Made semiconductor data from 20+ client companies queryable** — including Samsung, SK and LG — by designing a SQL database prototype managing **1M+ entities** in Oracle and SQLite
- **Cut retrieval time across large-scale datasets** by restructuring and optimizing the queries the team relied on

PROJECTS

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the GLSL pipeline I owned: PCF shadows, screen-space reflections, vertex AO, day-night cycle

Passenger | Unreal Engine 5, HLSL, Maya

Mar 2026 – Apr 2026

- **Gave a short film its entire visual identity** — **2 HLSL post-process shaders** (anisotropic Kuwahara, cell shader) over GBuffer channels, delivered as a ray-traced 2560×1080 render
- **Brought its character to life** — rigged and animated in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- **Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops (split, extrusion, triangulation) in a Qt editor

TECHNICAL SKILLS

Graphics & Rendering: OpenGL, GLSL, HLSL, shader programming, Unreal Engine 5, Lumen, real-time & offline rendering

3D & Tooling: Maya, Blender, Substance Painter, Qt, Movie Render Pipeline

Languages: C++, Python, TypeScript, JavaScript, Java, C#, GLSL/HLSL, SQL

Frameworks & Tools: React Three Fiber, Three.js, React, Node.js, Git, GitLab, VS Code